Areas Related to Circles

The big idea

Real objects often use **parts** of circles: a slice of pizza (sector), the region a wiper cleans, a window shaped like a segment. We find their areas by taking a **fraction** of the whole circle.

PARTS OF A CIRCLE

Sector: region between two radii and an arc (a pizza slice). **Segment**: region between a chord and its arc.
The smaller one is the **minor** sector or segment; the rest is **major**.

Arc length and sector area

A central angle θ is θ/360 of the full turn, so everything is that fraction of the full circle.

FOR A SECTOR OF ANGLE Θ IN A CIRCLE OF RADIUS R

$$\text{Length of arc} = (\theta/360) \times 2\pi r$$

$$\text{Area of sector} = (\theta/360) \times \pi r^2$$

FIND THE AREA OF A SECTOR OF ANGLE 60° IN A CIRCLE OF RADIUS 6 CM (Π = 3.14)

$(60/360) \times 3.14 \times 36 = (1/6) \times 113.04 =$ **18.84 cm²**.

A CLOCK'S MINUTE HAND IS 14 CM. FIND THE AREA IT SWEEPS IN 5 MINUTES.

In 60 minutes it turns 360°, so in 5 minutes it turns 30°.
Area = (30/360) × (22/7) × 14 × 14 = **154/3 ≈ 51.33 cm²**.

Area of a segment

AREA OF A MINOR SEGMENT

Area of segment = area of sector − area of the triangle formed by the two radii and the chord

For θ = 60° the triangle is equilateral (area (√3/4) r²). For θ = 90° it is a right triangle (area ½ r²). For θ = 120°, the triangle's area is (√3/4) r².

CHORD SUBTENDS 90° AT THE CENTRE OF A CIRCLE OF RADIUS 10 CM. FIND THE MINOR SEGMENT'S AREA (Π = 3.14).

Sector = (90/360) × 3.14 × 100 = 78.5 cm². Triangle = ½ × 10 × 10 = 50 cm².
Segment = **28.5 cm²**. Major segment = 314 − 28.5 = **285.5 cm²**.

USING THE DIAMETER AS R

Every formula uses the **radius**. If the question gives the diameter, halve it first.

SHADED-REGION QUESTIONS

Shaded area = (big shape) − (unshaded shapes). Write this word equation first, then fill in numbers.

Quick recap

- Arc = (θ/360) 2πr. Sector = (θ/360) πr².
- Segment = sector − triangle.

- Major part = whole circle – minor part.

Practice questions

1. The area of a quadrant of a circle of radius 14 cm is ($\pi = 22/7$)
   (a) 616 cm² (b) 154 cm² (c) 308 cm² (d) 77 cm²
2. The length of an arc of angle 90° in a circle of radius 7 cm is
   (a) 22 cm (b) 11 cm (c) 44 cm (d) 5.5 cm
3. A sector of a circle of radius r has angle 180°. Its area is
   (a) $\pi r^2/2$ (b) πr^2 (c) $\pi r^2/4$ (d) $2\pi r^2$
4. Area of segment equals
   (a) sector + triangle (b) sector – triangle (c) triangle – sector (d) circle – sector
5. In 5 minutes a minute hand turns through
   (a) 5° (b) 6° (c) 30° (d) 60°

Answers: 1. b 2. b 3. a 4. b 5. c